Green University of Bangladesh

CSE 403 – Class Test 1 – **Complete Solution**

(a) Weights w_1 (slope) and w_0 (intercept) [2 marks]

Formula:

$$w_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}, \qquad w_0 = \bar{y} - w_1\bar{x}$$

Step 1: Compute means

$$\bar{x} = \frac{2+3+4+5+6}{5} = 4 \qquad \bar{y} = \frac{28+35+42+48+55}{5} = 41.6$$

Step 2: Compute summations

$$\sum(x_i - \bar{x})(y_i - \bar{y}) = 67.0 \qquad \sum(x_i - \bar{x})^2 = 10$$

Step 3: Compute slope

$$w_1 = \frac{67.0}{10} = 6.7$$

Step 4: Compute intercept

$$w_0 = 41.6 - (6.7 \times 4) = 41.6 - 26.8 = 14.8$$

Final Answer:

$$\hat{y} = 14.8 + 6.7x$$

(b) Prediction at $x = 4.5$ [1 mark]

Formula:

$$\hat{y} = w_0 + w_1 x$$

Substitution:

$$\hat{y} = 14.8 + (6.7 \times 4.5)$$

Solution:

$$\hat{y} = 14.8 + 30.15 = 44.95$$

Final Answer: $\hat{y} \approx 44.95$

(c) Performance Metrics [3 marks]

MAE Formula:

$$MAE = \frac{1}{n}\sum|y_i - \hat{y}_i|$$

MAE Calculation:

$$MAE = \frac{0.2+0.1+0.4+0.3+0.0}{5} = 0.2$$

RMSE Formula:

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

RMSE Calculation:

$$RMSE = \sqrt{\frac{0.30}{5}} = \sqrt{0.06} \approx 0.245$$

Residual Sum of Squares ($\mathbf{SS_{res}}$) Formula:

$$SS_{res} = \sum(y_i - \hat{y}_i)^2$$

Step-by-step calculation:

$$SS_{res} = (0.2)^2 + (0.1)^2 + (0.4)^2 + (0.3)^2 + (0.0)^2$$

$$SS_{res} = 0.04 + 0.01 + 0.16 + 0.09 + 0.00 = 0.30$$

Total Sum of Squares ($\mathbf{SS_{tot}}$) Formula:

$$SS_{tot} = \sum (y_i - \bar{y})^2$$

Step 1: Mean

$$\bar{y} = 41.6$$

Step 2: Deviations

$$SS_{tot} = (28 - 41.6)^2 + (35 - 41.6)^2 + (42 - 41.6)^2 + (48 - 41.6)^2 + (55 - 41.6)^2$$

Step 3: Expand

$$SS_{tot} = (-13.6)^2 + (-6.6)^2 + (0.4)^2 + (6.4)^2 + (13.4)^2$$

Step 4: Compute

$$SS_{tot} = 184.96 + 43.56 + 0.16 + 40.96 + 179.56 = 449.2$$

R^2 **Formula:**

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Substitution:

$$R^2 = 1 - \frac{0.30}{449.2}$$

Solution:

$$R^2 \approx 0.9993$$

Final Interpretation: The model explains more than 99.9% of the variance in the data.